

Power Consumption (Dual motor) 5 Minutes

- A continuous 5-minute (311 seconds) diagnostic test was conducted to evaluate the dual-motor setup. The microcontroller was programmed to supply a capped maximum PWM signal (180/255) to safely regulate the 14.8V source to the motors' 12V rating.
- To simulate a real-world agricultural field deployment where the **rover encounters uneven terrain** (e.g., one wheel stuck in mud while the other spins freely), mechanical friction was applied asymmetrically. **Motor 2 was subjected to heavy physical resistance**, while **Motor 1 operated near a zero-resistance state**. Current draw (mA) was calculated continuously via the BTS7960 shunt resistors and exported via serial monitoring every 1,000 milliseconds.

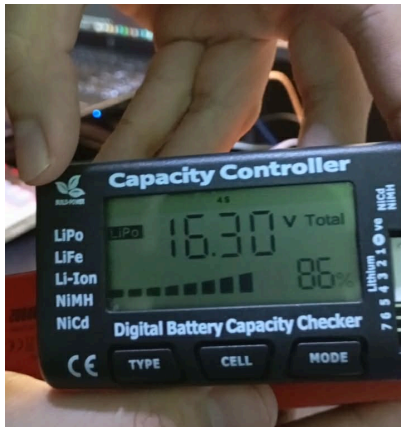
Results & Data Logging

Serial Monitor Output (Sample at the 5-Minute Mark, T=297s to T=311s)

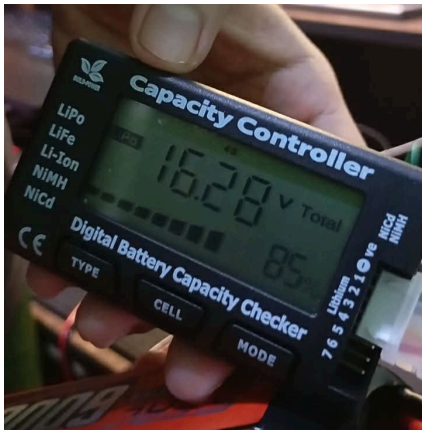
Timestamp (s)	Motor 1 (Free-Spin) mA	Motor 2 (Loaded) mA	Total System mA	
296	37.40	789.66		827.07
297	0.00	793.82		793.82
298	8.31	0.00	8.31	
299	16.62	1051.50		1068.12
300	24.94	174.56		199.49
301	16.62	935.12		951.75
302	54.03	739.79		793.82
303	20.78	0.00	20.78	
304	12.47	0.00	12.47	
305	58.19	523.67		581.86
306	29.09	423.92		453.02
307	20.78	926.81		947.59
308	78.97	856.16		935.12
309	66.50	0.00	66.50	
310	0.00	610.95		610.95
311	24.94	407.30		432.24

Battery Capacity

- **Pre-Test Battery State:** 16.30V (86% Capacity)



- **Post-Test Battery State:** 16.28V (85% Capacity)



- **Total Measured Drain:** 0.02V drop (1% Capacity utilized)

Data Analysis

System Component	Recorded Metric	Conclusion
Load Distribution	Motor 1: ~10-50 mA Motor 2: 800-1,150 mA	Successfully simulated an asymmetric traction-loss event (free-spin vs. heavy-terrain friction).
Peak Current	1,292.55 mA (1.29 A)	The locomotion subsystem is mathematically proven to be electrically stable under stress.
BTS7960 Drivers	Operating at < 3% of 43A capacity	Excellent thermal safety; drivers effortlessly handled the simulated mechanical resistance.
Voltage Stability	0.02V drop over 5 minutes	Demonstrated exceptional voltage stability across the 14.8V power supply.
Capacity Drain	~60 mAh (1% of 6000mAh)	Primary battery holds ample capacity for extended field missions; a secondary drive battery is unnecessary.